Huawei Access Node WHI-ACC-50 QoS Congestion Event Resolution SOP

Node: WHI-ACC-50 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_5129

Description

This Standard Operating Procedure (SOP) addresses the failure domain of QoS congestion events, specifically microbursts, interface tail drops, and QoS queue starvation on Huawei Access Node WHI-ACC-50 located at the Whitefield East POP. This failure domain is primarily caused by the interaction of QoS policies, interface configurations, and scheduling mechanisms of the Node's Access Control List (ACL) and Quality of Service (QoS) features. The underlying protocol mechanics involve the allocation of bandwidth and buffer resources to packets based on their DSCP (Differentiated Services Code Point) values, which can lead to congestion when not properly configured. Inadequate ACLs and QoS policies can cause packets to be dropped or delayed, resulting in microbursts, interface tail drops, and QoS queue starvation. Additionally, hardware limitations, such as buffer size constraints, can further exacerbate the issue. Therefore, this SOP outlines a step-by-step approach to diagnose and resolve such congestion events on Huawei Access Node WHI-ACC-50, ensuring optimal network performance and QoS.

Pre-requisite Checks

- Pre-requisite: Verify that the Node WHI-ACC-50 is reachable and accessible via the Huawei CLI., Description: Ensure that the remote hands are dispatched to the Whitefield East POP to access the Node.

- Pre-requisite: Check the status of redundant links to ensure they are operational., Description: Verify that the redundant links are up and functioning correctly to prevent any single point of failure.

Potential Root Causes

Cause	Probability
Inadequate QoS policies causing QoS queue starvation.	High
Insufficient ACL configurations leading to packet drops.	Medium
Hardware limitations, such as buffer size constraints.	Low

Diagnostic Commands & Expected Outputs

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Command: display qos policy, Expected Output: Policy name: Policy-1, Class name: Class-1, Description: This command displays the current QoS policy and class configurations, helping to identify any potential issues with policy settings.

Command: display acl {acl-number}, Expected Output: Access Control List {acl-number} entries: 10, Description: This command displays the ACL entries, enabling the identification of potential packet drops or
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delays due to inadequate ACL configurations.

Command: display interface {interface-name} queue, Expected Output: Queue: 1, Drop Count: 1000, Description: This command displays the interface queue statistics, including the drop count, which can indicate potential congestion issues.
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Resolution Steps

Step 1: Step: Adjust QoS policies to prevent queue starvation., Command: config qos policy {policy-name} class {class-name} bandwidth {bandwidth-value}, Description: Adjust the QoS policy to allocate sufficient bandwidth to prevent queue starvation.

Step 2: Step: Reconfigure ACLs to prevent packet drops., Command: config acl {acl-number} entry {entry-number} permit {source-ip} {source-port} {destination-ip} {destination-port}, Description: Reconfigure the ACL to permit packet forwarding and prevent packet drops.

Step 3: Step: Increase buffer size to alleviate congestion., Command: config interface {interface-name} queue buffer-size {buffer-size-value}, Description: Increase the buffer size to alleviate congestion and prevent packet drops.

Post-Check Verification

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Command: display qos policy, Expected Output: Policy name: Policy-1, Class name: Class-1, Bandwidth: {bandwidth-value}, Description: Verify that the QoS policy has been adjusted to prevent queue starvation.

Command: display acl {acl-number}, Expected Output: Access Control List {acl-number} entries: 10, Permit: {source-ip} {source-port} {destination-ip} {destination-port}, Description: Verify that the ACL has been reconfigured to prevent packet drops.

Command: display interface {interface-name} queue, Expected Output: Queue: 1, Drop Count: 0, Description: Verify that the interface queue statistics have improved, indicating a resolution to the congestion issue.
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Roll-back Procedures

- Step: Rollback QoS policy changes., Command: undo config qos policy {policy-name} class {class-name} bandwidth {bandwidth-value}, Description: Rollback the QoS policy changes to the previous configuration.

- Step: Rollback ACL changes., Command: undo config acl {acl-number} entry {entry-number} permit {source-ip} {source-port} {destination-ip} {destination-port}, Description: Rollback the ACL changes to the previous configuration.

- Step: Rollback buffer size changes., Command: undo config interface {interface-name} queue buffer-size {buffer-size-value}, Description: Rollback the buffer size changes to the previous configuration.